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Disentanglement based Active Learning

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Abstract

We propose Disentanglement based Active Learning (DAL), a new active learning technique based on query synthesis which leverages the concept of disentanglement. Instead of requesting labels from the human oracle, our method automatically labels majority of the datapoints, thus drastically reducing the human labelling budget in active learning. The proposed method uses Information Maximizing Generative Adversarial Nets (InfoGAN) to achieve the task where the active learner provides a feedback on the generation of InfoGAN based on which decision is taken about the datapoints to be queried. Results on two benchmark datasets demonstrate that DAL is able to achieve nearly fully supervised accuracy with fairly less labelling budget compared to existing active learning approaches.

1 Introduction

Large amount of labelled data is required to train deep learning models for classification tasks. However gathering labels for training instances is expensive and time-consuming. Active learning tries to overcome this bottleneck by asking an oracle (for e.g., a human annotator) to label selected unlabelled instances out of the pool, thus achieving very high accuracy using as few labelled instances as possible[21]. The samples to be added to the training set in each active learning cycle are chosen by a query strategy based on the active learner which is continued until we obtain the required accuracy or reach a fixed labelling budget. This approach is known as pool based active learning.

The success of machine learning algorithm at its heart lies in the quality of the representation/features learned from data. The features learned by current deep learning models have various limitations. Current machine learning models lack understanding of concepts. They mainly rely upon statistical irregularities for achieving their desired task [7] and are vulnerable to adversarial attacks [24]. Disentangled representation is a representation in which single latent units are sensitive to changes in single generative factors, while being relatively invariant to changes in other factors[2]. InfoGAN is a variant of Generative Adversarial Nets (GAN) which learns disentangled representations in an unsupervised manner. It has been proven to be quite successful in disentangling representations such as digit identity from digit shapes in MNIST datasets, background images from the central digits in SVHN dataset etc.[3]. We do not use variants of GANs such as Conditional Generative Adversarial Nets [16], Auxiliary Classifier GANs [17] which employ label conditioning because we do not have enough labelled instances to train these architectures in an active learning setting.

In this paper, we mainly focus on active learning using disentanglement. The proposed method aims at reducing human supervision for active learning with the help of InfoGAN. The samples for each active learning cycle are synthesized by an InfoGAN out of which some of the datapoints are given to the human oracle to label based on the feedback from the active learner. Our approach ensures that very few datapoints are labelled by the annotator.

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Our contributions are as follows :

- We propose a new approach DAL for active learning which significantly reduces the human labelling budget.
- We propose a novel query strategy which incorporates a simple feedback mechanism from active learner along with uncertainty sampling.
- We perform extensive set of experiments using two different benchmark datasets (MNIST, CIFAR- 2 classes) and discuss the limitations of our approach and possible solutions.
- To the best of our knowledge, this is the first active learning framework which uses the concept of disentanglement.

2 Related work

Active Learning. Active learning is a common approach in many modern machine learning problems where we have a large pool of unlabelled data but labels are expensive to obtain[21]. Various criteria are used to select datapoints from the pool, the most common being uncertainty sampling.

Application of GAN in active learning. Active learning using GAN (GAAL) is first proposed in [27] where uncertain samples are generated by GAN instead of selecting samples from the pool in each iteration of active learning. One of the main limitations of GAAL is that the images generated from GANs might not be human interpretable. Our approach generates samples using InfoGAN for active learning but incorporates a mechanism which reduces the labelling budget and reduces the burden on human annotators which makes it different from GAAL. We also show that our method outperforms GAAL in terms of average human labelling budget. [11] proposed an active learning method using Conditional GAN (CGAN) specifically for medical image segmentation. There are pool based active learning approaches which make use of GAN. For instance, [15] generates the samples based on uncertainty sampling using GAN and nearest neighbours of synthesised instances from the unlabelled pool are added to the training data in each active learning cycle.

Disentangled representation learning. Recently, extensive experiments have been conducted in the field of disentangled representation learning for generative modelling. InfoGAN is a variant of GAN which learns interpretable and disentangled representations in an unsupervised manner[3]. There are other variants of GANs such as Structured GAN (SGAN) [4] and Triple GAN [14] which relies on semi supervised conditional generation to improve disentanglement. Variational Autoencoders have been extended to learn disentangled representations in [6].

3 Disentanglement based Active Learning

We propose a new novel approach to reduce the human annotation cost in GAN based active learning approaches leveraging the property of disentanglement in InfoGAN. The main advantage of our approach is that it significantly reduces the labelling burden of humans compared to the other GAN based active learning approaches and achieves nearly fully supervised accuracy. One other advantage is that our approach generates images uniformly from all the classes in each active learning cycle. One cycle of DAL is depicted in Figure 1.

3.1 Generation of uncertain samples using InfoGAN

The main aim of InfoGAN is to learn interpretable and meaningful representations from the images in an unsupervised setting. [3] proposes a simple modification to the loss function of GAN to learn disentangled representations. This is achieved by adding a regularization term to the objective function of GAN. The input to the InfoGAN is a noise vector(z) sampled from a simple distribution as in ordinary GAN along with a set of latent codes($c_1, c_2, ..., c_n$) which has meaningful effects on the output. Let c be the concatenated vector of all the vectors c_i where $i = 1, 2, .., L$. [3] assumes that the latent codes are independent and hence is given by $P(c_1, c_2, .., c_n) = \prod_{i=1}^{L} P(c_i)$. Training is done as in original GANs but also maximizes the Mutual Information between the latent codes and

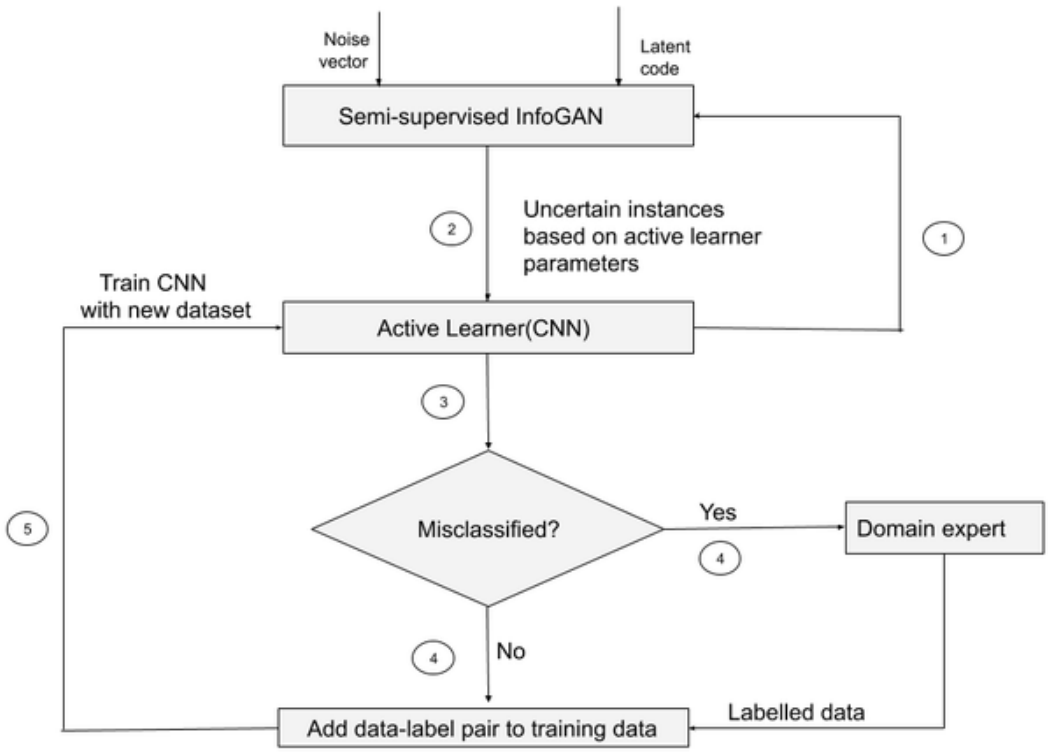

Figure 1: Working of InfoGAN based Active Learning

the generator output which eventually maps latent code to meaningful representations. The objective function for InfoGAN is given by :

$$\min_{G} \max_{D} V_I(D, G) = V(D, G) - \lambda I(c; G(z, c)) \tag{1}$$

where $V(D, G)$ is the objective function of original GAN, $I(c; G(z, c)$ is the mutual information between latent code c and the generator output $G(z, c)$. λ is hyperparameter which is set to 1. Since it is impractical to calculate mutual information explicitly, the real distribution $P(c|x)$ is approximated to an auxillary distribution $Q(c|x)$ which is modeled by a parameterized neural network.

The peculiarity of InfoGAN or any architecture that learns discrete disentangled representations is that it also learns to disentangle class categories. Once the architecture is trained and it learns interpretable representation, it gives us an advantage of generating instances from the required class since the discrete latent code maps to class category. This property is utilized in DAL. InfoGAN is trained beforehand and uncertain instances are generated from generator G in each active learning cycle by minimizing the optimization problem as in ASAL[15]. Uncertainty is computed for each generated sample using information entropy as in [8]. We use information entropy because it is suitable for both binary and multiclass classification problems. The samples with maximum entropy are considered as the most informative samples.This can be formulated as an optimization problem which is given by

$$z^* = \arg\min_{z} \sum_{m \in \mathcal{C}} p_{\theta(k)}(m|G(z, c)) \log p_{\theta(k)}(m|G(z, c)). \tag{2}$$

The noise vector alone in InfoGAN is minimized by stopping the gradient flow to the latent codes, hence leaving it unchanged. Once we get z^*, uncertain samples are generated by passing it through the generator G of InfoGAN. In each active learning cycle, the discrete latent codes are initialised in such a way that the InfoGAN generates samples uniformly from all the class categories. This ensures that the datapoints added to the training dataset in each active learning cycle are uniformly

distributed from all the classes and are uncertain. Ideally, all the generated samples can be added to the training set for the next iteration as in GAAL since the class information is already obtained from the discrete latent codes. However, the disentangling can't be completely trusted since InfoGAN can be erroneous in generating datapoints.

3.2 Feedback generation from active learner

InfoGAN has an error rate of 5% in its generation on MNIST dataset[3]. So, the key idea in our approach to alleviate this problem is to use the knowledge acquired by the active learner till this point of time to decide if the generated samples from InfoGAN are from the corresponding classes mapped by the discrete latent codes. Otherwise, the samples are provided to the human experts to label. We use a simple strategy to handle the generation error in InfoGAN. The samples generated from InfoGAN are passed through the current active learner to detect misclassifications in each active learning iteration. The misclassified samples are given to domain experts for labelling. In case of correct classification by active learner, we assume that the generated samples are from the same class category on which latent code is mapped to and thus there is no need of labelling. For example, the latent code corresponding to class 9 is provided as the input to InfoGAN. Let the image generated be 7. If the generated image is forward passed through the current learner and it predicts class 7, the latent code of InfoGAN is not in agreement with the prediction of active learner. Hence, the data is provided to the domain expert to label. Otherwise, InfoGAN has accurately generated image corresponding to the latent code and hence there is no need to label which is directly added to the training set along with the class label. This approach is thus an effective way of reducing the number of datapoints to be labelled by a domain expert.

4 Experiments

We conduct experiments on two different datasets, MNIST and CIFAR-10. We perform multiclass classification on MNIST dataset and binary classification on CIFAR-10 dataset by extracting two classes, horse and automobile similar to [27]. This is mainly because human annotators can easily distinguish between horse and automobile classes. Even though our approach requires less human intervention, we chose horse and automobile so as to ensure fair comparison with GAAL. We denote CIFAR-2 for binary classification on horse and automobile classes extracted from CIFAR-10 to avoid confusion. We also implement and compare our approach to four different schemes that are listed as follows :

- GAN based Active learning (GAAL) from [27].
- Random sampling using GAN, where random instances synthesized by GAN are provided to the oracle to label.
- Uncertainty sampling from pool, where uncertain instances are selected from the unlabelled pool[13].
- Fully supervised learning, where all samples in the pool are used to train the classifier.

The architecture of InfoGAN used for the proposed approach is described in [3]. We use all the unlabelled instances in the pool to train InfoGAN so that it learns better representations. Since the InfoGAN architecture used is very similar to that of DCGAN, we use DCGAN architecture[18] for all the implementations that use GAN to ensure fair comparison with our approach. We use LeNet[12] with cross entropy loss as classifier for all the implementations. Early stopping (patience = 10) is employed to prevent overfitting. The maximum number of epochs is set to 100. The model is trained with Adam optimizer[10] with learning rate 0.001.

Stochastic gradient descent is used to optimize Equation (2). The gradient update is done for a batch of noise vectors in each active learning cycle instead of minimizing each z vector in a sequential manner. Minimisation is done for 100 steps where the learning rate is set to 0.001. We initialise the training data with 100 datapoints that are randomly sampled from the unlabelled pool. 50 samples are added to the training data in each active learning cycle. This process is repeated until the size of training data is 10000. The experiments are performed for 5 different random seeds. The classification accuracies for all the above schemes are reported on the original MNIST and CIFAR-2 testing data.

5 Results

We perform experiments to analyse the classification accuracies and average labelling budget required for different active learning strategies listed above and compare it with the proposed approach.

The fully supervised accuracy achieved by Lenet-5 on MNIST dataset is 99.5%. It can be observed in Figure 2 (Top) that the proposed approach achieves an accuracy of 97.2%(+/-0.2) with just 10,000 datapoints. A similar tendency can be seen on CIFAR-2 binary class classification. Also, the accuracy curve of DAL is steeper compared to the other algorithms for initial few cycles indicating that the queried points generated by our criteria is much more informative compared to that of the other algorithms. This indicates the effectiveness of the query strategy used in DAL. It is interesting to note that none of the GAN based approaches achieve accuracy comparable to the pool based active learning setting. We suspect this is because the instances added to the training dataset are selected from the pool of unlabelled data in pool based uncertainty sampling whereas they are synthesized using GAN in case of GAN based approaches. The generated instances might not follow the exact same distribution as that of the actual data [1] which makes the training and testing distribution different.

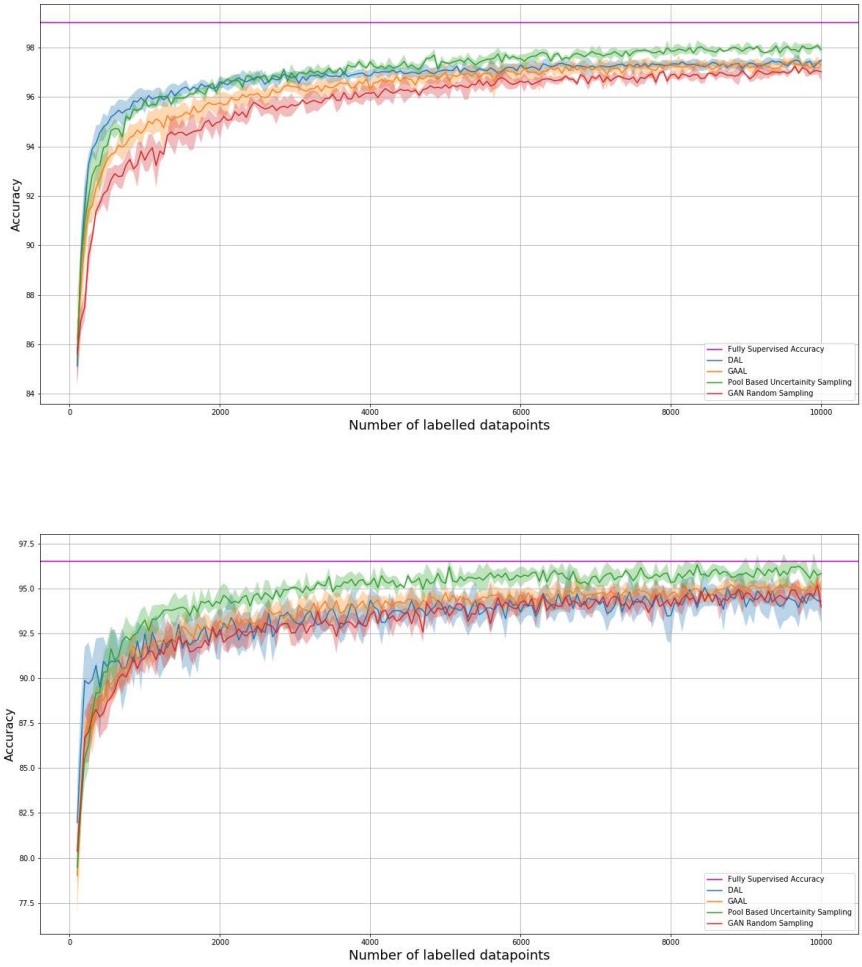

Figure 2: (Top) Validation accuracy on *MNIST* dataset for various active learning strategies. (Bottom) Validation accuracy on *CIFAR-2* for various active learning strategies.

It can also be observed that in Figure 2 (Bottom), GAAL performs better than DAL on CIFAR-2. However, this behavior is not stressed on MNIST dataset. This is possibly due to lower GAN-Train

measure [22] of InfoGAN compared to DCGAN whose effect is much more evident on a complicated dataset like CIFAR-2. The mutual information term incorporated in InfoGAN loss function restricts its capacity to learn the actual data distribution and hence performs worse compared to DCGAN. This problem can be eliminated by incorporating better architecture like Wasserstein GAN[20] which approximates the data distribution well into InfoGAN architecture as experimented in [26].

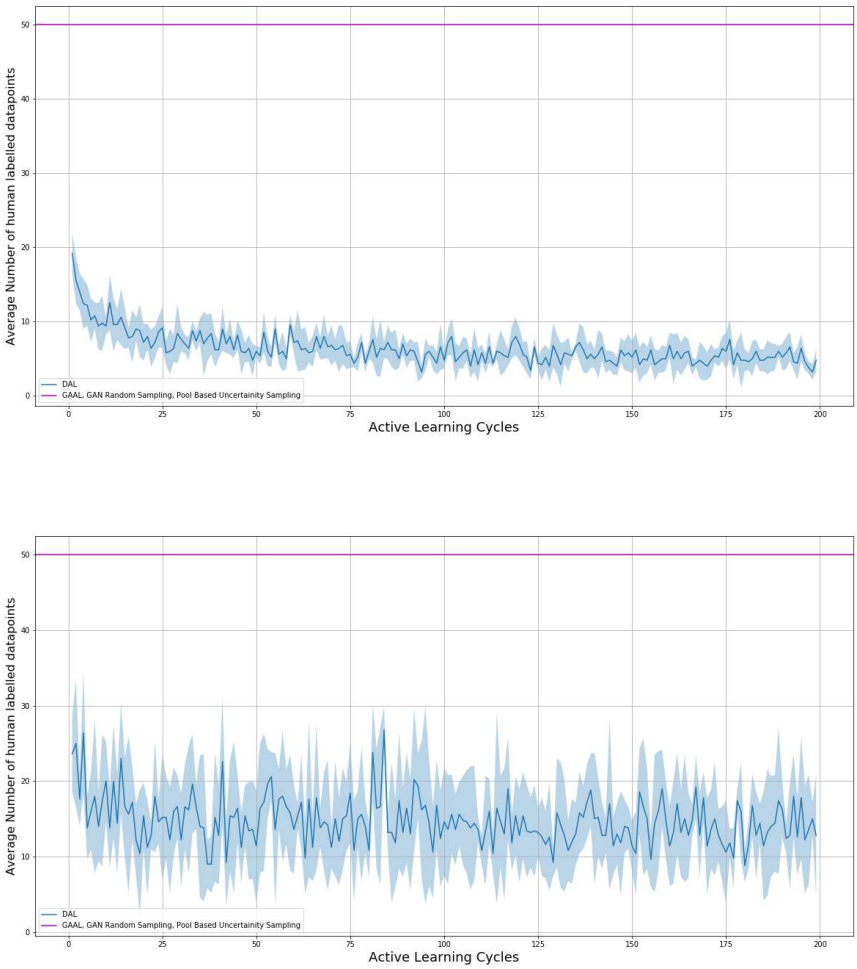

Figure 3: (Top) Average labelling budget on *MNIST* dataset for various active learning strategies. (Bottom) Average labelling budget on *CIFAR-2* for various active learning strategies.

The effectiveness of DAL can be clearly observed in Figure 3. As mentioned earlier, 50 datapoints are added to the training dataset in each active learning cycle. In the other 3 approaches except DAL, all the 50 datapoints are to be labelled by the domain experts and hence 10,000 datapoints are to be labelled in total in both the datasets. On the contrary, the average number of points to be labelled in DAL is less than 20 till 25 learning cycles while it is even less after 25 active learning cycles. An instance is provided to human expert for labelling only if the active learner predicts a class different from that of the class mapped to the actual latent code of the data generated in InfoGAN . It can be noticed that the average labelling budget in DAL decreases from that of the initial cycles or both the datasets. As the active learner learns better and better over the cycles, it becomes better in classification task and hence reports only if the InfoGAN and active learner disagree. Thus the average number of labelled datapoints is only around 1400 and 2940 out of 10000 in MNIST and CIFAR-2 respectively which is a drastic reduction of annotation cost compared to the other approaches.

6 Conclusion and Future work

We proposed a novel approach for the reduction of human labelling budget in active learning by integrating the concept of disentanglement along with GAN based active learning techniques and a simple yet effective feedback mechanism from the active learner. One limitation of the proposed approach is that it blindly depends on the feedback of the active learner to decide if the data should be annotated by domain experts. There are chances of addition of noisy instances to the training data which adversely affects the training paradigm. This problem can be alleviated by making active learner robust to noisy labels [5],[19]. [9] performed label-noise robust training on GANs. We suspect incorporating such techniques in InfoGAN helps us to eliminate the problem of noisy labels. One other possible solution is to improve disentanglement of class categories by adding semi supervision[23] or using better architectures like SGAN, Triple-GAN so that less noisy instances are added to the training set. We also explore the idea of jointly training the InfoGAN and active learner as in [25] which seems to be an interesting direction.

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